NASCAR SPONSORSHIP VISIBILITY ANALYSIS

Final Report — 2024 Season

Strategic Findings & Sponsor Recommendations

Client	Analyst	Date
NY Racing	Anshul Ghildiyal	2026

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Executive Summary

Situation. NY Racing is evaluating primary and secondary NASCAR Cup Series sponsorship opportunities for the 2026 season across five incumbent sponsors spanning three team tiers — from elite (Hendrick Motorsports) to lower-tier (Legacy Motor Club, Front Row Motorsports).

Complication. Sponsorship decisions in motorsport have historically been driven by relationships and intuition rather than quantitative visibility data. Without objective cross-channel metrics, the team cannot fairly compare sponsorship value across vastly different cost tiers, or identify which partnerships deliver the strongest return on investment.

Question. Which of the five tracked sponsors delivered the greatest visibility exposure during the 2024 season? And which — adjusted for estimated sponsorship cost — delivers the best visibility per dollar invested?

Answer. Our composite visibility scoring model — built across 36 races and five sponsors using race performance, media, social, and event data — produces a clear, actionable picture. Monster Energy is the primary recommendation, offering the best balance of high visibility (#2, 1,343 pts) and strong efficiency (#3, 76.8 pts/$M). Fastenal is the secondary recommendation for ROI-first mandates (77.5 pts/$M). Love's Travel Stops is the budget-conscious alternative (82.0 pts/$M, #1 efficiency).

Headline Statistics

Metric	Value
Sponsors analysed	5
Races tracked	36 (full 2024 Cup Series season)
Correlation pairs tested	8 of 8 significant at $p < 0.0001$
Win visibility multiplier	~9x versus average finish
NAPA efficiency disadvantage vs. leader	32% — pays elite premium for marginal visibility gain
Visibility ranking stability	Perfectly stable across all 4 weight sensitivity scenarios

Key Findings

Finding 1 — Laps Led outperforms Finish Position on every visibility channel. Laps Led correlates with Reddit at r = +0.871, News at r = +0.864, and YouTube at r = +0.854 — consistently 0.24–0.29 points higher than Finish Position. On-screen dominance mid-race drives more coverage than the final result.

Finding 2 — Race wins generate a non-linear ~9x visibility multiplier. A win is not simply "a very good finish." It triggers disproportionate coverage across all three channels simultaneously. NAPA and Monster Energy (4 wins each) both cleared 1,300 season points; Fastenal's single win provided a meaningful mid-tier lift above its expected average.

Finding 3 — Elite-team sponsors pay a visibility premium that compresses ROI. NAPA Auto Parts leads raw visibility (1,363 pts) but ranks last in efficiency (62.0 pts/$M) due to its ~$22M estimated cost. Love's Travel Stops, at ~$5.5M, achieves 82.0 pts/$M — 32% more efficient per dollar.

Finding 4 — All rankings are perfectly stable across weight sensitivity tests. Whether race performance is weighted at 25% or 50%, or all categories receive equal weight, the five sponsors always rank in exactly the same order. The model's conclusions are robust to reasonable weight assumptions.

Top-Line Results

Sponsor	Vis. Score	Vis. Rank	Efficiency	Eff. Rank	Est. Cost	Wins
NAPA Auto Parts	1,363	#1	62.0 pts/$M	#5	$22M	4
Monster Energy	1,343	#2	76.8 pts/$M	#3	$17.5M	4
Fastenal	853	#3	77.5 pts/$M	#2	$11M	1
Love's Travel Stops	451	#4	82.0 pts/$M	#1	$5.5M	0
Dollar Tree	410	#5	68.4 pts/$M	#4	$6M	0

Theoretical max: 3,600 pts. Efficiency = season visibility score / estimated annual cost ($M). All 8 correlation pairs are significant at $p < 0.0001$.

Section 3: Analysis Findings

3.1 EDA Insights — Performance-Visibility Correlations

Before building the scoring model, we tested eight performance-visibility relationships across the full 180-record dataset (36 races x 5 sponsors). All eight returned p-values below 0.0001, confirming that on-track results reliably drive sponsorship exposure across every channel.

Relationship	r	Strength	Channel
Laps Led vs Reddit Mentions	+0.871	Strong	Social
Laps Led vs News Mentions	+0.864	Strong	Media
Laps Led vs YouTube Sponsor Views	+0.854	Strong	Video
Laps Led vs YouTube View Share	+0.762	Strong	Video
Finish Position vs YT View Share	-0.699	Mod-Strong	Video
Finish Position vs YouTube Views	-0.615	Moderate	Video
Finish Position vs News Mentions	-0.585	Moderate	Media
Finish Position vs Reddit Mentions	-0.579	Moderate	Social

Table 3.1 — All 8 correlations significant at $p < 0.0001$. Laps Led outperforms Finish Position by 0.24–0.29 r-points on every channel.

Key Surprise: Laps Led dominates Finish Position on every channel. Finish position was assumed to be the primary driver going in. Instead, Laps Led produced correlations 0.24–0.29 points higher. The explanation is mechanical — leading laps means sustained camera time at the front of the field for hours; a race finish is a single late-stage event. This finding directly elevated Laps Led's weight in the composite model.

Additional EDA Highlights

Finding	Direction
Win bonus is non-linear — approximately 9x the average visibility score	Unexpected magnitude
Top-5 finishes drive 10.75x more Reddit volume than P11+ finishes	Threshold effect
Playoff races average 27.7% LOWER visibility than regular season (H3 confirmed)	Counter-intuitive
Dollar Tree shows no finish-Reddit correlation (r = -0.145, p = 0.40)	Brand floor issue

Table 3.2 — Notable EDA findings. The playoff finding (H3) is flagged for discussion with Sarah Mitchell before weight finalisation.

3.2 Visibility Score Results

The composite visibility model assigns each sponsor a weekly score (0-100 scale, weighted across four categories) and sums all 36 races into a season total (theoretical max: 3,600 pts). Season-wide min-max normalisation ensures cross-sponsor comparability. All 5 sponsors have full 36-race datasets.

Sponsor	Season Score	Avg/Race	Wins	Avg Finish	Top-5 %	Laps Led
NAPA Auto Parts	1,363	37.9	4	10.0	33.3%	770
Monster Energy	1,343	37.3	4	9.2	38.9%	641
Fastenal	853	23.7	1	16.0	11.1%	144
Love's Travel Stops	451	12.5	0	25.7	0.0%	0
Dollar Tree	410	11.4	0	27.6	0.0%	0

Table 3.3 — Monster Energy's better average finish (9.2) is offset by NAPA's higher laps-led count (770 vs 641), explaining NAPA's narrow 20-point lead.

Model Validation — 6 of 6 checks passed, 0 anomalies: finish-visibility correlation r = -0.987; win-visibility r = +0.983; laps-visibility r = +0.970; win/non-win score ratio 2.04x; playoff impact -27.7% (H3 confirmed); top-5 vs bottom-20 ratio 2.8x.

3.3 Efficiency Analysis

Efficiency = season visibility score / estimated annual sponsorship cost ($M). Cost estimates are derived from Forbes, Sports Business Journal, and industry benchmarks. All three cost scenarios (low/mid/high) were tested for sensitivity.

Sponsor	Vis. Score	Eff. Rank	Efficiency	Cost/Point	Cost Range	Cost Sensitivity
Love's Travel Stops	451	#1	82.0 pts/$M	$12,191	$3.5M – $7.5M	±3 rank swing
Fastenal	853	#2	77.5 pts/$M	$12,902	$8M – $14M	0 rank swing
Monster Energy	1,343	#3	76.8 pts/$M	$13,028	$14M – $21M	±3 rank swing
Dollar Tree	410	#4	68.4 pts/$M	$14,622	$4M – $8.5M	±2 rank swing
NAPA Auto Parts	1,363	#5	62.0 pts/$M	$16,137	$18M – $26M	±1 rank swing

Table 3.4 — Efficiency rankings (mid-cost scenario). Fastenal is the most cost-stable (0-rank swing). Monster Energy is the most sensitive — always present with cost range in stakeholder discussions.

Key Headline: NAPA Auto Parts leads raw visibility but pays an elite-team premium that places it last in efficiency. Sponsors can achieve 82.0 pts/$M (Love's), 77.5 pts/$M (Fastenal), or 76.8 pts/$M (Monster Energy) — all dramatically more efficient per dollar than NAPA's 62.0 pts/$M.

3.4 Sensitivity Analysis

Scenario	Performance	Media	Social	Events	Visibility Ranking
Baseline (used)	40%	30%	20%	10%	NAPA > Monster > Fastenal > Loves > Dollar — Perfectly Stable
Performance Heavy	50%	25%	15%	10%	NAPA > Monster > Fastenal > Loves > Dollar — Perfectly Stable
Media Heavy	30%	40%	20%	10%	NAPA > Monster > Fastenal > Loves > Dollar — Perfectly Stable
Equal Weights	25%	25%	25%	25%	NAPA > Monster > Fastenal > Loves > Dollar — Perfectly Stable

Table 3.5 — All 5 visibility rankings are perfectly stable across all 4 weight scenarios — a strong indicator of model robustness.

Section 4: Recommendations

Based on a multi-criteria evaluation matrix (40% visibility / 40% efficiency / 20% additional factors), three sponsors are recommended for NY Racing's sponsorship strategy, each serving a distinct stakeholder priority.

PRIMARY RECOMMENDATION

MONSTER ENERGY — Tyler Reddick / 23XI Racing / #45

Best Overall Balance

Metric	Monster Energy	Dataset Avg	Context
Visibility Score	1,343 pts	882 pts	#2 of 5 — 98.5% of leader
Efficiency	76.8 pts/$M	73.3 pts/$M	#3 of 5 — only 7% below leader
Estimated Cost	$17.5M/yr	—	Competitive-tier (medium confidence)
Wins	4	1.8	4x wins = sustained top-tier engagement
Avg Finish	9.2	17.7	Best average finish among all 5 sponsors
Top-5 Rate	38.9%	16.7%	Highest top-5 rate in dataset
Laps Led	641	341	88% of NAPA at a fraction of the cost gap
Reddit Mentions	3,716	1,752	2.1x dataset average — active community link

Why Monster Energy?

- Best combined rank across both dimensions. Monster ranks #2 in visibility and #3 in efficiency — combined rank 5, tied for the lowest composite in the dataset. It is the only sponsor defensible to both the CMO (near-maximum visibility) and the CFO (above-average efficiency).
- Best on-track performance metrics in the full dataset. A 9.2 average finish and 38.9% top-5 rate outperform even Chase Elliott's NAPA-sponsored #9 car. Four race wins mean four ~9x visibility spikes through the season.
- Efficiency within 7% of the leader at 3x the absolute visibility. Love's Travel Stops leads efficiency (82.0 pts/$M) but delivers only 451 pts. Monster Energy achieves 76.8 pts/$M while generating 1,343 pts — nearly 3x more raw exposure for a marginal efficiency trade-off.
- Active fan community link confirmed. With 3,716 Reddit mentions — 2.1x the dataset average — the brand's NASCAR association resonates with the r/NASCAR community independently of raw race outcomes.

Risk Assessment — Monster Energy

Risk	Likelihood	Impact	Mitigation
Cost estimates carry +/-$3.5M uncertainty	Medium	Eff. rank could drop to #4	Negotiate cost cap; retest efficiency at each quote
23XI charter dispute (2024, races 20-28)	Low ongoing	Data anomaly in 9 races	Review races 20-28 individually before signing
2025 SHR closure reshapes competitive field	Certain	Moderate upside likely	Model is 2024 baseline; validate with 2025 data
Single-driver concentration risk	Low	Visibility collapse if DNF run	Seek multi-race performance clauses in contract

SECONDARY RECOMMENDATION

FASTENAL — Chris Buescher / RFK Racing / #17

Best ROI Play

Metric	Fastenal	Dataset Avg	Context
Visibility Score	853 pts	882 pts	#3 of 5 — solid mid-tier performance
Efficiency	77.5 pts/$M	73.3 pts/$M	#2 of 5 — most cost-stable rank (0-rank swing)
Estimated Cost	$11M/yr	—	Mid-tier — half of Monster Energy
Wins	1	1.8	One win generates meaningful spike across channels
Avg Finish	16.0	17.7	Slightly above dataset average
Cost Sensitivity	0-rank swing	—	Rank #2 in low, mid, and high cost scenarios

Why Fastenal as Secondary?

- Most defensible efficiency story for ROI-mandated stakeholders. Fastenal's #2 efficiency rank holds across ALL low, mid, and high cost scenarios with zero rank movement. If the CFO demands hard ROI evidence, Fastenal is the only sponsor where efficiency can be quoted without caveat.
- Accessible entry price at a mid-tier team. At $11M/year, Fastenal costs half of Monster Energy and less than a quarter of NAPA's premium — a viable path for budget-disciplined mandates that can still be presented as data-backed.
- Buescher's 2024 win demonstrates breakout capability. RFK Racing is a competitive mid-tier team with realistic probability of repeated top-5 appearances, offering upside visibility that is not priced into the mid-tier estimate.
- Trade-off vs Monster Energy: Fastenal delivers 77.5 vs 76.8 pts/$M — marginally more efficient. But absolute visibility is 853 vs 1,343 pts (37% less exposure). The choice is fundamentally a budget question, not an efficiency question.

BUDGET-CONSCIOUS ALTERNATIVE

LOVE'S TRAVEL STOPS — Todd Gilliland / FRM / #34

Love's Travel Stops leads the dataset in efficiency at 82.0 pts/$M — 32% better than NAPA and 7% better than Fastenal. At ~$5.5M/year, it represents the lowest-barrier entry into NASCAR primary sponsorship in this analysis. Best suited for NY Racing if budget constraints are the binding priority, or as a test-and-validate partnership before scaling to a higher-tier sponsor.

Metric	Love's Travel Stops	Context
Visibility Score	451 pts	#4 of 5 — lower tier, data-verified
Efficiency	82.0 pts/$M	#1 of 5 — dataset leader
Estimated Cost	$5.5M/yr	Lowest commitment in analysis
Wins / Top-5	0 / 0	Performance risk — no breakout events in 2024
Best-case efficiency	128.9 pts/$M	If actual cost is at low estimate ($3.5M)
Cost Sensitivity	3-rank swing	Rises to #3 efficiency at high cost assumption

Critical Caveat — H5 Lower-Tier Floor Effect: Love's and Dollar Tree operate at a performance tier where the r/NASCAR community does not track race-by-race outcomes for these brands (Dollar Tree: $r = -0.145$, $p = 0.40$). Their visibility is driven by baseline brand presence, not performance feedback loops. Improving Gilliland's results may not translate linearly into more visibility — a key structural limitation distinguishing lower-tier from elite or mid-tier sponsors. A minimum-visibility floor adjustment (H5) is recommended in model v1.1.

4.4 Strategic Scenario Comparison

Three scenarios map directly to different stakeholder priority profiles. No single scenario dominates all dimensions — the choice depends on NY Racing's budget flexibility and board-level objectives.

	Scenario A	Scenario B	Scenario C
Label	Maximize Visibility	Maximize Efficiency	Balanced
Best For	CMO / brand growth mandate	CFO / ROI mandate	Board consensus / mixed priorities
Sponsor	NAPA Auto Parts	Love's Travel Stops	Monster Energy
Visibility	1,363 pts (100%)	451 pts (33%)	1,343 pts (98.5%)
Efficiency	62.0 pts/$M (76%)	82.0 pts/$M (100%)	76.8 pts/$M (94%)
Est. Cost	$22M/year	$5.5M/year	$17.5M/year
Vis vs Max	100% — maximum	33% — sacrifice	98.5% — near-maximum
Eff vs Max	76% — sacrifice	100% — maximum	94% — near-maximum
Key Trade-off	Max exposure but lowest ROI	Best ROI but lowest exposure	Near-max on both; not #1 on either
Choose When	Budget not a constraint	Hard budget cap or ROI test	Optimise across stakeholder goals

Table 4.2 — Scenario C (Monster Energy) achieves 98.5% of maximum visibility and 94% of maximum efficiency — the most attractive combined position in the dataset.

NY Racing Recommendation: For NY Racing's 2026 planning, Scenario C (Monster Energy) is the recommended path. It delivers near-maximum visibility at a cost 20% below NAPA, with an efficiency profile only 7% below the dataset leader. This is the only scenario that generates a compelling narrative for both the CMO (visibility story) and the CFO (ROI story) without meaningful sacrifice on either axis. If budget is cut below ~$15M, pivot to Scenario B (Fastenal) - not Scenario A.

4.5 Implementation Guidance & Next Steps

Step	Action	Owner	Timeline
1	Share model and efficiency findings with Sarah Mitchell and Marcus. Flag H3 (playoff weight) and Monster cost range for discussion.	Anshul	Week 7
2	Obtain actual cost quotes from 23XI Racing and RFK Racing to replace estimates; recalculate efficiency with verified figures.	NY Racing Partnership team	Week 7-8
3	Validate Monster Energy races 20-28 for 23XI charter dispute anomalies; recalculate if outliers detected.	Anshul	Week 7
4	Extend analysis to 2022-2023 seasons for multi-year trend validation before finalising recommendation.	Anshul	Post Week 8
5	Build model v1.1 with H5 lower-tier floor adjustment and updated playoff weight (pending Sarah sign-off on H3).	Anshul	Post Week 8
6	Conduct brand-fit qualitative assessment (audience alignment, activation value, hospitality) to supplement quantitative scores.	NY Racing + Extern PM	Concurrent

Table 4.3 — Steps 1-3 are critical path before any sponsorship negotiation begins.

Decision Framework for Future Evaluations

Step	Action	Output
1	Collect race performance data (36-race season)	Race results dataset
2	Gather social and media data (Reddit, YouTube, News)	Platform engagement dataset
3	Run 7-variable composite scoring model	Normalised weekly visibility scores
4	Research sponsorship cost estimates in three scenarios	Low / mid / high cost table
5	Calculate efficiency (visibility pts / cost $M)	Efficiency ranking table
6	Run weight and cost sensitivity tests	Stability assessment
7	Prepare scenario comparison document	Stakeholder decision framework

Appendix A: Methodology Summary

A.1 Model Variables & Weights

Variable	Category	Eff. Weight	Direction	Transform
finish_position	Race Performance	28%	Inverse	None
laps_led	Race Performance	12%	Direct	None
news_weighted_mentions	Media Coverage	30%	Direct	None
reddit_mentions	Social Engagement	10%	Direct	None
youtube_sponsor_views	Social Engagement	10%	Direct	log(x+1)
is_win	Special Events	6%	Binary	x100
is_playoff	Special Events	4%	Binary	x100

Table A.1 — 5 variables excluded (redundant or volatile): reddit_total_score, reddit_engagement_per_mention, youtube_total_views, youtube_view_share, news_total_mentions.

A.2 Data Sources

Data Type	Source	Coverage	Confidence
Race Results	Kaggle NASCAR 2024 + Racing Reference	36 races, 5 sponsors	High
Reddit Data	r/NASCAR via PRAW API (public JSON)	36 race weeks	Medium-High
YouTube Data	YouTube Data API v3	36 race weeks	Medium-High
News Mentions	Google News (manual, tiered weighting)	36 race weeks	Medium
Sponsorship Costs	Forbes, Sports Business Journal, HardHead Marketing	Estimates only	Low-Medium

Table A.2 — Sponsorship costs are estimates. All efficiency findings should be re-run with verified cost quotes before negotiation.

A.3 Known Limitations

Limitation	Impact	Mitigation
Cost estimates +/-20-30% uncertainty	Efficiency ranks could shift +/-1-3 positions	Test with low/mid/high; obtain actual quotes
Single-season scope (2024 only)	Cannot confirm multi-year trend stability	Extend to 2022-2023 in model v1.1
Proxy metrics (not direct impressions)	Captures correlates of visibility, not exposure itself	Consistent with Zoomph industry methodology
H3: Playoff weight retained at 4%	Slight overweight given negative finding	Discuss with Sarah Mitchell; adjust in v1.1
H5: Lower-tier floor not yet applied	Dollar Tree / Love's may be underestimated	Add baseline offset in model v1.1
23XI charter dispute (races 20-28)	Possible data anomaly for Monster Energy	Manually review those 9 races

Table A.3 — Known limitations and mitigations.

A.4 File Reference

File	Description	Status
master_dataset.csv	180 rows x 23 cols — core season data	Complete
visibility_scoring_model.ipynb	Full Python scoring implementation	Complete
scored_dataset.csv	180 rows + weekly visibility scores	Complete
efficiency_rankings.csv	Efficiency ranks with low/mid/high cost sensitivity	Complete
validation_report.md	6/6 model validation checks + H1-H5 hypothesis results	Complete
methodology.md	313-line model specification document	Complete
NASCAR_Master_Dataset_and_Dictionary.docx	Full data dictionary	Complete
NASCAR_Correlation_Findings.docx	Week 4 EDA correlation report	Complete